P1284R0
Allowing Inlining of Replaceable Functions

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Abstract

Allow replaceable functions to be inlined

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§ 1. Motivation

For many allocations (and deallocations, when using sized delete), it would be profitable to inline the implementation.

- The practical issues encountered due to the One Definition rule are discussed by [P1220].
- Under 15.5.4.6 [replacement.functions], the ability to inline the replaceable functions used for allocation is forbidden ("The program’s declarations shall not be specified as inline. No diagnostic is required.").

This paper focuses on the ability to specify replaceable functions as inline. This restriction came about as the resolution to [LWG404], adopted at the Sydney meeting [N1629].

§ 2. Proposal

Wording is relative to [N4762].

- 15.5.4.6 [replacement.functions]

> ~~The program’s declarations shall not be specified as inline. No diagnostic is required.~~

§ References

§ Informative References

[LWG404]
May a replacement function be declared inline?. 2016-12-23. URL: http://cplusplus.github.io/LWG/lwg-defects.html#404

[N1629]
Minutes of J16 Meeting No. 38/WG21 Meeting No. 33, March 22-26, 2004. 2004-04-08. URL: http://www.open-std.org/jtc1/sc22/wg21/docs/papers/2004/n1629.html

[N4762]
Working Draft, Standard for Programming Language C++. 2018-07-07. URL: http://www.open-std.org/jtc1/sc22/wg21/docs/papers/2018/n4762.pdf

[P1220]
Controlling When Inline Functions are Emitted. 2018-10-05. URL: https://wg21.link/P1220